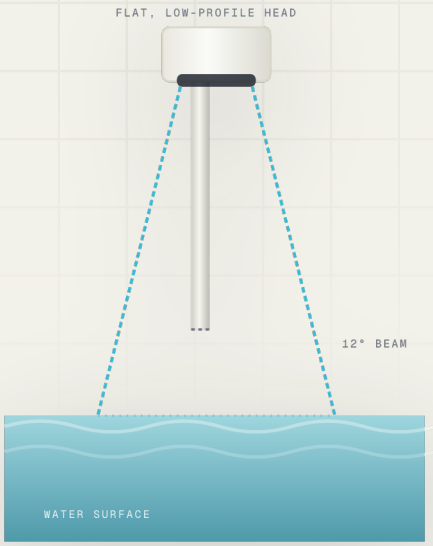
VEPOLINK
MONITORING PURITY

PROCESS INSTRUMENTATION × WATER LEVEL

NON-CONTACT WATER LEVEL, YEARS IN THE FIELD

PulseRadar *Level Sensor*

A flat, low-profile pulse radar sensor that measures water level without ever touching the water — unaffected by temperature, humidity, floating debris or contaminated flow, and built for the lowest power budgets.



0.4–35 m
MEASUREMENT RANGE

±3 mm
BEST-CASE ACCURACY, 2-30M

<24 mA
ACTIVE CURRENT DRAW

NON-CONTACT PULSE RADAR | **SOLAR-READY** ULTRA-LOW POWER | **VEPOLINK** FIELD-PROVEN BUILD

0contact
SENSOR MOUNTS ABOVE THE WATER

<0.05mA
RESTING CURRENT, SOLAR-READY

12°
BEAM WIDTH, NARROW & FOCUSED

2–20sec
MEASUREMENT TIME PER READING

01 OVERVIEW

Water level, *read from the air.*

A non-contact pulse radar sensor for long-term surface water measurement. A flat, low-profile antenna sends short radar pulses down to the water and times the return, with nothing ever touching the flow — no fouling, no corrosion, no moving parts to wear.

Extremely low power consumption makes it ideal for remote or solar-powered sites, and the light, compact body mounts easily on a bridge, an extension bracket, or inside a small protective housing.

REPORTED DATA

- WATER LEVEL / DISTANCE TO WATER
- SDI-12 OUTPUT
- RS-485 (VIA SDI-12)
- 4-20 MA, FIELD-SCALABLE
- INTERNAL DATA LOGGER

· PRODUCT FEATURES

Flat, Low-Profile Design

An unobtrusive flat antenna eliminates nesting areas for insects and reduces periodic maintenance, while staying unobtrusive at urban or vandalism-prone sites.

Flexible Integration

Connects to most dataloggers via SDI-12, RS-485 (using SDI-12), or 4–20 mA — no additional PC software needed for setup.

Low Power Consumption

Requires only around 24 mA when active and under 0.05 mA at rest — ideal for remote or solar-powered monitoring sites.

High Performance in Any Condition

Measurements are unaffected by air temperature, humidity, flood events, floating debris, or contaminated water, reducing missed data and post-processing.

02 MEASUREMENT PRINCIPLE

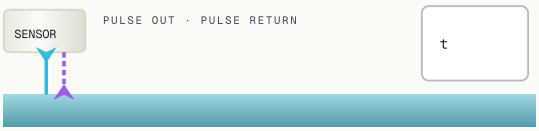
How level is read *without touching water.*

The sensor emits short radar pulses downward toward the water surface and measures the time for each pulse to reflect back. That time-of-flight converts directly into distance to water, and from there into water level above a known datum.

PULSE RADAR

Time-of-Flight Ranging

SHORT PULSE · REFLECTION · ELAPSED TIME



A short radar pulse travels down to the water surface and reflects back; the elapsed time gives distance to water with no contact at any point.

`t ∝ distance`

ANTENNA

Flat, Narrow-Beam Head

12° (±6°) BEAM WIDTH · FLAT PROFILE



A narrow 12° beam keeps the measurement focused on the water directly below, while the flat housing avoids insect nesting and stays low-visibility on site.

`Focused, not scattered`

OUTPUT

Standard Interfaces

SDI-12 · RS-485 · 4-20 MA, SCALABLE



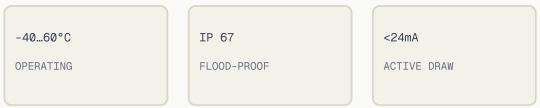
The sensor connects to most dataloggers and telemetry platforms directly, with the 4–20 mA output scalable to a smaller measuring range in the field.

`No extra software needed`

ENVELOPE

Built for the Field

-40 TO 60 °C · IP67 · LOW POWER



A wide operating temperature range, IP67 flood-proof housing, and very low active current suit unattended sites of any kind.

`Remote & solar-ready`

03 SIGNAL & INTERFACE

Three ways in, *one reading out.*

Whatever the logger or telemetry platform on site already speaks, the sensor speaks it too — without a dedicated configuration tool.

INTERFACE 01

SDI-12

Configure via the SDI-12 transparent mode of a datalogger, or a USB/SDI-12 adapter, with no separate PC software required.

INTERFACE 02

RS-485 (using SDI-12)

Runs the same SDI-12 command set over an RS-485 line for longer cable runs — up to 100 m — back to the logger.

INTERFACE 03

4–20 mA

A conventional analog loop output, scalable in the field to a smaller measuring range to match an existing control system.

CABLE RUNS

The sensor supports cable lengths up to 100 m back to the logger or control panel, making it practical to mount the sensor on a bridge or pier well away from the enclosure that reads it.

WHY NON-CONTACT MATTERS

Contact sensors in open water face fouling, corrosion and debris strikes over time. Because the pulse radar sensor never touches the water, none of those failure modes apply — the measurement stays reliable through floods, sediment loads and seasonal changes alike.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

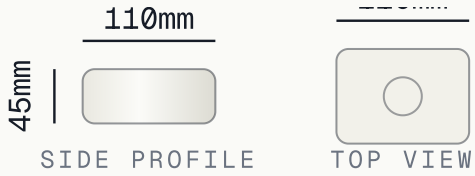
Parameter	Water level / distance to water
Working Principle	Non-contact pulse radar
Measurement Range	0.4 - 35 m
Beam Width	12° (±6°)
Measurement Time	2 - 20 sec
Accuracy, 0.4–2 m	± 10 mm
Accuracy, 2–30 m	± 3 mm
Accuracy, 30–35 m	± 10 mm
Accuracy, 4–20 mA Output	± 0.1% full scale
Resolution	1 mm · 0.1 mbar

SYSTEM · BUILD · POWER

Interface	SDI-12 · RS-485 · 4-20 mA
Internal Data Logger	Yes
Operating Frequency	25.3 GHz
Transmission Power	5 mW
Power Supply	5.4 - 28 V DC, typ. 12/24 V DC
Power Consumption, Active	< 290 mW (< 24 mA at 12 V)
Power Consumption, Resting	< 0.6 mW (< 0.05 mA at 12 V)
Housing Material	ASA (UV-stabilised ABS)
Protection Class	IP 67
Operating Temperature	-40 to 60 °C
Storage Temperature	-40 to 85 °C
Max. Cable Length	100 m

SENSOR HEAD DIMENSIONS

INDICATIVE SIDE & TOP · MM



05 INSTALLATION & SITING

Mounted above, *reading below.*

> Mount the sensor above the highest water level expected at the site, aimed straight down, with the beam cone clear of any structure, cable or obstruction. No wiring to the water itself is required — only power and signal back to the logger, over cable up to **100 m.**

01 · MOUNTING HEIGHT

Above max flood stage

Mount with clearance above the highest water level the site will ever see, respecting the sensor's near-range blanking distance.

02 · BEAM CLEARANCE

Clear 12° cone

Keep the narrow beam cone free of bridge piers, rails, cables or other obstructions between the sensor and the water.

03 · SURFACE CONDITIONS

Avoid turbulence

Choose a stretch of water that is as smooth as possible within the beam footprint, away from inflow turbulence or wave action.

04 · MOUNTING HARDWARE

Bridge, bracket or housing

Fits a bridge deck, an extension mounting bracket, or inside a small protective housing depending on the site.

05 · WIRING

SDI-12, RS-485 or 4–20 mA

Wire directly into the site's existing datalogger or telemetry platform using whichever interface it already supports.

06 · COMMISSIONING

Datum & range check

Confirm the zero datum against a staff gauge reading and verify the measured range covers the site's full flood stage.

OBSTRUCTION CHECKING

Check the beam path at the lowest anticipated water level for submerged obstructions such as rocks or bridge piers that could disturb the reading once water rises to cover them.

MAINTENANCE

With no wetted parts and a flat antenna that resists insect nesting, maintenance is limited to periodic visual checks of the mounting and beam path — there is nothing submerged to clean or replace.

06 APPLICATIONS

Where it *works.*

Any surface water site where a sensor cannot sit in the water itself — because of debris, contamination, vandalism risk, or simply the cost of maintaining a wetted sensor over years of service.

Streams & Rivers	Tidal Zones	Reservoirs & Lakes
Flood Warning	Sediment & Debris-Laden Channels	Bridges & Culverts

MIGRATING CHANNELS

Because the measurement is taken from above rather than at a fixed submerged point, it tolerates a channel bed that shifts over time far better than a contact sensor anchored in the bed itself.

URBAN & VANDALISM-PRONE SITES

The flat, low-profile housing draws less attention than a projecting sensor, which matters at urban drainage points and other sites exposed to the public.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Pulse radar time-of-flight: the sensor sends a short radar pulse down to the water and measures the time for the reflection to return.

Q Does the sensor touch the water?

No. It mounts above the water and measures entirely from the air, with no wetted parts at all.

Q What range does it cover?

0.4 to 35 m, with accuracy as tight as ±3 mm across the 2–30 m mid-range band.

Q How much power does it use?

Under 24 mA active and under 0.05 mA at rest — low enough for small solar panels or long-life battery supplies.

Q Does floating debris or contamination affect the reading?

No — being non-contact, the sensor is unaffected by air temperature, humidity, flood events, floating debris or contaminated water.

Q What outputs are available?

SDI-12, RS-485 (using SDI-12), and a field-scalable 4–20 mA output.

Q Is separate configuration software needed?

No — configure via the SDI-12 transparent mode of a datalogger or a USB/SDI-12 adapter.

Q How long can the cable run be?

Up to 100 m between the sensor and the logger or telemetry panel.

Q What is the beam width?

12° (±6°) — narrow enough to keep the measurement focused on the water directly below the mount.

Q What maintenance does it need?

Minimal — periodic checks of the mounting and beam path. There are no wetted parts to clean, foul or replace.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

WATER LEVEL, READ WITHOUT TOUCHING IT

Know your level. *No contact required.*

Tell us the site's range, mounting point and power source, and we'll confirm the fit and configure the output. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that sizes, specifies and commissions non-contact level sensors. No call centre.
+91 99538 86178 →

02 **Request a site sizing** RECOMMENDED
Send your expected range, mounting height and power source — we'll confirm the fit and output.
sales@vepolink.com →

03 **Book a demonstration**
See a running sensor and its output with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM